Making NDFU bottom-up: the unified adoption rule and its two variants

NDFU internal design note

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Status: design document — no changes to the engine have been made.
Companion scripts: `10_unified_model_sweep.R`, `12_unified_premium_sweep.R`

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1 Why this document

The Social Networks special-issue rejection (Bianchi's follow-up email) said:

> *"the underdevelopment of the behavioural model assumed in your ABM. While it can certainly be considered an agent-based model, the behaviour of the agents is relatively*

abstract, and the experimental design appears to be governed by macro-level manipulation rather than bottom-up dynamics."

Two pieces of the current model are exposed to that critique:

1. **The network threshold τ_i is exogenous.** It is drawn from $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$, truncated to $[0, 1]$, *independently of the agent's own measured preference* q_i. The sweep over $(\tau_{\text{mean}}, \tau_{\text{sd}})$ is exactly the "macro-level manipulation" Bianchi points at: a population-level dial with no micro-foundation in the survey data.
2. **The two channels are bolted together.** Rational adoption ($q_i \leq \Gamma$) and social adoption ($\tau_i \leq \tilde{E}_i$) are two separate rules joined by an OR. A referee can ask why an agent runs two unrelated decision procedures.

Tur, Zeppini & Frenken (2024, *Social Networks* 76:12–21) — published in our target journal — has a *single* micro rule (a utility comparison, where social reinforcement enters the utility calculus), but **no social-distance sensitivity in influence**: every adopting neighbor reinforces equally. NDFU's distinctive ingredient is exactly that sensitivity (Centola 2011). The question is therefore:

> Can we keep NDFU's selective (distance-sensitized) influence while adopting a Tur-style single utility-based decision rule — making the model bottom-up without giving up its identity?

Answer: yes. The instrument is a single **unified adoption rule** (§4), which admits two operationalizations — a *linear-fractional* variant and a *multiplicative γ-variant* that nests Tur (2024) exactly. Their pros and cons are compared in detail in §5.

2 Notation and the current NDFU model (baseline)

Symbol	Meaning	Source
$\Gamma \in [0, 1]$	Intrinsic Utility Level (IUL) of the behavior	swept
$q_i \in [0, 1]$	Minimum Utility Requirement (MUR): intrinsic aversion	**empirical** (GSS/ATP items)
$\tau_i \in [0, 1]$	network threshold (fraction of influential adopters needed)	**exogenous** $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$
$d_{ij} \in [0, 1]$	Gower distance in Blau space (age, educ, race, relig, sex)	**empirical**
$h \in [0, 1]$	Maximum Social Proximity (MSP): tolerance to dissimilarity	swept (macro)
$a_j \in \{0, 1\}$	adoption state of j	dynamic
X_{ij}	adjacency	imputed ERGM networks

Current rules (as in the paper).

- Rational channel: adopt if $q_i \leq \Gamma$ (Eq. 1 of the paper).
- Social channel: adopt if $\tau_i \leq \tilde{E}_i$, with selective effective exposure

$$\tilde{E}_i \;=\; \frac{\sum_{j \neq i} X_{ij}\, \tilde{a}_j}{\sum_{j \neq i} X_{ij}}, \qquad \tilde{a}_j = 1 \;\iff\; a_j = 1 \wedge U_{ij} \leq \sigma(h - d_{ij}), \tag{1}$$

where $U_{ij} \sim \text{Unif}(0, 1)$ is redrawn per step and $\sigma(z) = \left(1 + e^{-z/T}\right)^{-1}$ with $T = 0.02$ in the implementation (a near-hard gate at $d_{ij} = h$; see `scripts/04_GSS_diffusion_sims.R`, line 304).

Implementation nuance worth knowing (it matters for the comparison with Tur): in the code, the rational channel is implemented by setting the node's threshold to 10^{-6}, so a rational agent still needs **one influential contact** (exposure > 0) before adopting. The engine is therefore already a word-of-mouth model for *awareness*: low-q agents behave as simple contagion, not as spontaneous ($\varphi = 0$) adopters. The paper's Eq. 1 (adoption at exposure 0) is the idealization; the code is contact-gated.

3 Tur (2024) mechanism, for reference

- Word-of-mouth **percolation**: agents must be *informed* through the network before they can decide.
- Single decision rule: adopt when the innovation's intrinsic utility ξ meets the agent's minimum utility requirement.
- **Social reinforcement modifies the requirement** — exact published form (confirmed against both PDFs):

$$u_t^i \;=\; u_0^i \cdot \left(\frac{1}{a_t^i}\right)^{\gamma}, \tag{2}$$

  where u_0^i is the initial minimum utility requirement, $a_t^i \in \mathbb{N}$ the number of **adopting neighbors** of i at time t (a state count, not received recommendations; the rule is evaluated only once informed, so $a \geq 1$), and $\gamma \geq 0$ the social-reinforcement strength. The lowering is **multiplicative** (power-law): each doubling of adopting neighbors divides the requirement by the same factor 2^{γ}. With $\gamma = 0$ the model reduces to pure social percolation. The threshold-like behavior emerges from this lowering, not from a separate threshold parameter.
- Homophily ρ is purely **structural pre-wiring**: agents with similar u are placed together *before* diffusion. Influence itself is **distance-blind**: any adopting neighbor reinforces equally.
- Preferences are artificial distributions (Beta(α, β) family in 2018; uniform in 2024); networks are Watts–Strogatz with $N = 10{,}000$ and $k = 4$ neighbors on average.

NDFU vs. Tur, in one line: **Tur has the single bottom-up rule but uniform influence; NDFU has selective influence but a two-channel rule with an exogenous τ.** Each has what the other lacks.

4 The unified additive-utility rule

4.1 The rule

Replace both channels by one utility comparison per agent per step:

$$\text{adopt}_i(t) = 1 \iff \Gamma + \lambda\, \tilde{E}_i(t) \;\geq\; q_i. \tag{3}$$

- $\tilde{E}_i(t)$ is **exactly the current selective effective exposure** (1), unchanged. This is where social distance enters: only adopters passing the stochastic proximity gate $\sigma(h - d_{ij})$ count.
- $\lambda > 0$ is the **social-proof coupling**: how much situational utility a fully-adopted, fully-similar neighborhood adds, in the same units as Γ and q_i.

4.2 Two algebraically identical framings (choose the second)

1. **Requirement-lowering (Tur's framing):** $q_i^{\text{eff}}(t) = q_i - \lambda \tilde{E}_i(t)$; adopt when $\Gamma \geq q_i^{\text{eff}}(t)$. Preferences are *modified* during diffusion.

2. **Situational-utility (recommended framing):** $U_i(t) = \Gamma + \lambda \tilde{E}_i(t)$; adopt when $U_i(t) \geq q_i$. The preference q_i **never changes**; what changes is the utility the situation offers (peers add legitimacy, lower uncertainty, network externalities). This is literally Granovetter's threshold as a *situated cost–benefit crossing* (1978, p. 1422), and it preserves NDFU's declared contrast with Tur ("preferences are not modified during diffusion", `theoretical_reviewed.md`, entry 7).

4.3 The derived threshold — the two old channels fall out as limit cases

Rearranging (3), adoption requires $\tilde{E}_i(t) \geq \tau_i^*$ with the **derived threshold**

$$\tau_i^* \;=\; \max\left\{0,\; \frac{q_i - \Gamma}{\lambda}\right\}. \tag{4}$$

- $q_i \leq \Gamma \;\Rightarrow\; \tau_i^* = 0$: adopts at first influential contact — **the current rational channel** (Granovetter's low-threshold "instigator", exactly as currently implemented with $\tau = 10^{-6}$).
- $q_i > \Gamma \;\Rightarrow\; \tau_i^* > 0$: needs selective exposure proportional to the **utility deficit** — the social channel, now derived instead of postulated.
- $q_i > \Gamma + \lambda \;\Rightarrow\; \tau_i^* > 1$: **can never adopt** ($\tilde{E} \leq 1$) — a "structurally immune" agent. When exactly this occurs is worked out in §5.1.

4.4 What this buys and what it costs

Buys.

- **Bottom-up microfoundation.** One decision rule, all heterogeneity empirical: τ_i^* is a deterministic function of the *measured* q_i and the product (Γ, λ). The arbitrary $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$ disappears. This is the direct answer to Bianchi.
- **Behavioral realism as a bonus:** resistance to peer influence is now positively correlated with intrinsic aversion (the same latent trait drives both) — defensible, and testable in principle.
- **Nesting.** Setting the gate to $w \equiv 1$ ($h \to \infty$, distance-blind) removes the distance selectivity and leaves a Tur-style (distance-blind) reinforcement model; setting $\lambda \to 0$ recovers a pure percolation-of-awareness model. Tur (2024)'s *exact* published form (2) is recovered by the γ-variant of the rule — see §4.6. NDFU's selectivity remains the *identified added ingredient*, cleanly positioned against a paper published in the target journal.
- **Implementability.** τ_i^* is static given (Γ, λ), so the existing `netdiffuseR` threshold machinery runs it as-is: `threshold.dist = pmax(0, (q - Γ)/λ)` replaces the Normal draw + the 10^{-6} hack. No engine changes needed.

Costs / changes.

- The rational/social decomposition becomes **ex-post bookkeeping** (adopters with $q_i \leq \Gamma$ vs. $q_i > \Gamma$) rather than two mechanisms — same tally we already compute, but the interpretation shifts.
- τ_{sd} disappears as a free dial; its large positive effect on cascades (heterogeneity creates susceptible minorities) is now carried by the *empirical* dispersion of q_i, scaled by $1/\lambda$. λ replaces $(\tau_{\text{mean}}, \tau_{\text{sd}})$ — the parameter space shrinks from 16 (4×4) threshold configurations to a 1-D λ sweep.
- The (IUL × MSP) heatmaps will not be identical: thresholds now co-vary with q. Whether the phase structure survives is an empirical question — answered: it does (see §7).

4.5 Where social distance enters (explicitly)

Only inside $\tilde{E}_i$, via the per-contact stochastic gate:

$$\Pr\big(j\text{'s adoption counts for } i\big) \;=\; \sigma\!\left(\frac{h - d_{ij}}{T}\right), \qquad T = 0.02. \tag{5}$$

Identical to the current model. The unified rule changes the *decision rule*, not the *influence filter*.

4.6 Recovering Tur (2024) exactly — the nesting, step by step

Tur's published rule (exact form, §3): an agent that has been *informed* through the network adopts when the product's utility meets its *current* requirement,

$$\xi \;\geq\; u_t^i, \qquad u_t^i = u_0^i \cdot \left(\frac{1}{a_t^i}\right)^{\gamma}, \tag{6}$$

with u_0^i the initial minimum utility requirement, $a_t^i \geq 1$ the number of adopting neighbors of i at time t (the rule is only evaluated once informed, so $a \geq 1$ and the expression is well-defined), and $\gamma \geq 0$ the social-reinforcement strength. $\gamma = 0$ gives $u_t = u_0$: pure social percolation (Solomon & Weisbuch 2000), Tur's own baseline.

Log-linearization — Tur IS an additive requirement-lowering rule, in logs. Taking logarithms (all quantities positive):

$$\ln \xi \;\geq\; \ln u_0^i \;-\; \gamma \,\ln a_t^i. \tag{7}$$

So the published reinforcement is additive with the concave gain $g(a) = \ln a$. This dictates the right unified rule: it must be additive in *log-utility*, i.e. **multiplicative in levels**.

The Tur-exact unified rule (γ-form). Define the distance-sensitized adopting *count*

$$\tilde{A}_i(t) \;=\; \sum_j X_{ij}\, \tilde{a}_j(t), \qquad \tilde{a}_j = 1 \;\iff\; a_j = 1 \,\wedge\, U_{ij} \leq \sigma\!\left(\frac{h - d_{ij}}{T}\right), \tag{8}$$

and let social proof *multiply* the situational utility:

$$\boxed{\text{adopt}_i(t) \;\iff\; \Gamma \cdot \big(\tilde{A}_i(t)\big)^{\gamma} \;\geq\; q_i} \qquad \text{(evaluated when } \tilde{A}_i \geq 1\text{)}. \tag{9}$$

Equivalently, in Tur's own requirement-lowering form: $q_{i,t} = q_i \cdot (1/\tilde{A}_i(t))^{\gamma}$, adopt when $\Gamma \geq q_{i,t}$. The preference q_i is fixed; what the similar adopters change is the *situational* requirement — the same framing as §4.2 (but see §5.5 for the substantive difference this form does carry).

Recovery, in one step. Switch off the proximity gate ($h \geq 1 + 5T \Rightarrow \sigma \to 1 \Rightarrow \tilde{a}_j = a_j$, so $\tilde{A}_i(t) = a_t^i$), and rule (9) reads

$$\Gamma \cdot (a_t^i)^{\gamma} \geq q_i \qquad \Longleftrightarrow \qquad \xi \;\geq\; u_0^i \cdot \left(\frac{1}{a_t^i}\right)^{\gamma} \tag{10}$$

under the identification

$$\boxed{\xi = \Gamma, \qquad u_0^i = q_i, \qquad \text{same } \gamma\,.} \tag{11}$$

The recovery is **exact on any graph**: Tur's reinforcement is a pure count, and the γ-form uses the count directly — no degree normalization is involved, so no degree-regularity caveat is needed (unlike the linear-fractional variant, which normalizes by degree and matches Tur's count reading only on degree-regular substrates).

Awareness / percolation layer (unchanged): the engine's contact-gating (the $\tau = 10^{-6}$ device, §2) means the rule is only evaluated at $\tilde{A}_i \geq 1$ — precisely Tur's "informed" condition with per-contact transmission probability 1; a probability $p < 1$ is emulated by a constant-acceptance gate ($\sigma \equiv p$).

Derived threshold (count form). Rearranging (9):

$$\text{adopt}_i \iff \tilde{A}_i(t) \geq c_i^*, \qquad c_i^* = \left(\frac{q_i}{\Gamma}\right)^{1/\gamma}. \tag{12}$$

- $q_i \leq \Gamma \Rightarrow c_i^* \leq 1$: adopts at the first influential contact — the rational channel, exactly as in the linear variant (§4.3).
- $q_i > \Gamma \Rightarrow$ needs c_i^* similar adopters; heterogeneity 100% empirical (from q_i), with γ as the single coupling.
- **Degree-dependent immunity**: agent i is out of reach whenever $c_i^* > k_i$ (its degree) — high-aversion, *low-degree* agents can never be socially converted (see §5.1).

What does NOT nest (honesty box).

- Tur's homophily ρ sorts agents by the requirement u *itself*; NDFU's imputation sorts by demographics, and q_i correlates with demographics only partially. The imputed networks are therefore an *empirically disciplined, weaker* version of Tur's preference-sorting. The nesting claim concerns the **adoption rule**; the network is an exogenous input in both models.

One-line summary. With the proximity gate switched off, the γ-form **is** Tur (2024) — same functional form, same γ, with $\xi = \Gamma$ and $u_0 = q$ — on any topology; NDFU's added ingredient is exactly one: the distance gate deciding *which* adopters count in $\tilde{A}_i$.

5 Pros and cons of the two operationalizations

We now have two members of one family on the table:

$$\textbf{Linear-fractional:} \quad \Gamma + \lambda\, \tilde{E}_i(t) \geq q_i \quad \text{(additive in the adopted } \textit{fraction}\text{)} \tag{13}$$

$$\gamma\textbf{-form (Tur-exact):} \quad \Gamma \cdot \tilde{A}_i(t)^{\gamma} \geq q_i \quad \text{(multiplicative in the adopted } \textit{count}\text{)} \tag{14}$$

This section works through the five questions that decide between them, pedagogically and one at a time, and closes with a verdict table.

5.1 When can an agent never adopt? (immunity, made precise)

In the linear variant, immunity is $\tau_i^* > 1$, i.e.

$$q_i > \Gamma + \lambda. \tag{15}$$

Since $q_i \leq 1$, this region is **non-empty only when** $\lambda < 1 - \Gamma$: with $\lambda \geq 1$ immunity is impossible (a fully-adopted similar neighborhood can compensate any deficit). So the intuition "$q_i, \Gamma \in [0, 1]$ makes this very unlikely" is correct *at large* λ but not in general. Empirically, on the GSS q distribution (diagnostic printed by `10_unified_model_sweep.R`):

λ	Γ	immunity condition	share immune
0.5	0.20	$q_i > 0.70$	10.9%
0.5	0.30	$q_i > 0.80$	3.8%
1.0–2.0	any	$q_i > \Gamma + \lambda \geq 1$	0%
0.7 (best fit)	0.20	$q_i > 0.90$	small tail

The visible fingerprint: the adoption **ceiling** $\Phi \approx 0.75$ at low IUL in the $\lambda = 0.5$ heatmaps — the $\approx 11\%$ immune tail (plus isolates) caps the ordered phase below 1. This is a prediction, not a bug: *when social proof is weak, its reach is bounded*. The current engine cannot produce this ceiling because τ is truncated to $[0, 1]$ by construction.

In the γ-form, immunity is instead $c_i^* > k_i$, i.e.

$$\left(\frac{q_i}{\Gamma}\right)^{1/\gamma} > k_i \quad \Longleftrightarrow \quad q_i \; > \; \Gamma\, k_i^{\gamma}, \tag{16}$$

which depends on the agent's **degree**: the out-of-reach agents are those both highly averse *and* poorly connected. Arguably more realistic — and a distinctive, testable difference between the two variants.

5.2 The log-space bridge: $\ln a$ vs. $\tilde{E}$

In log space, Tur's reinforcement variable is $\ln a$ — the log of the gated *count*. Is that "just" our normalized exposure? Almost: with $\tilde{E}_i = \tilde{A}_i / k_i$,

$$\ln \tilde{A}_i \; = \; \ln \tilde{E}_i \; + \; \ln k_i. \tag{17}$$

On **degree-regular substrates** — and Tur's Watts–Strogatz networks are degree-homogeneous with $k = 4$ — the term $\ln k_i$ is one global constant, absorbable into Γ (or u_0): identifying the reinforcement with $\ln \tilde{E}$ is then exact up to recalibration. On NDFU's **heterogeneous** networks (degrees roughly 1–80), the two differ by the agent-specific offset $\gamma \ln k_i$:

- using $\tilde{E}$ says "reinforcement is a *share of your attention*";
- using $\tilde{A}$ says "every endorsement adds the same, whether you have 4 neighbors or 40" — at the same adopted *fraction*, a hub receives more reinforcement.

This is the count-vs-fraction fork of §4.6, seen from the logarithm: the analogy $\ln a \leftrightarrow \tilde{E}$ is complete *modulo degree*, and the choice between the two variants is precisely the choice of what to do with $\ln k_i$.

5.3 What γ is — and is not — in Tur 2018 and Tur 2024

Read against both PDFs, the epistemic status of γ is the following.

What their justification does have.

- An **axiomatic characterization** (2018, p. 3; restated in 2024): the updated requirement $f(q; a, \gamma) = q \cdot (1/a)^{\gamma}$ must satisfy (a) decreasing in a for $\gamma > 0$; (b) decreasing in γ for $a > 1$; (c) $\gamma = 0$ replicates the percolation benchmark; (d) $a = 1$ gives $f = q$ (the *first* adopting neighbor only informs; reinforcement starts with the second — a clean echo of complex contagion's "more than one source").
- A **concavity argument** for the restriction $\gamma \leq 1$: "assuming that the marginal effect of each additional neighbor on the total social influence cannot be increasing" (2018), with the reductio that a very large γ trivializes diffusion *on their degree-4 lattices*.
- An interpretable **marginal effect** (2018, Eq. 2): $\Delta q_t^i = -\, q_0^i \gamma \,/\, (a_t^i)^{\gamma+1} \cdot \Delta a$ — the first adopting friends produce the large drops, and the drop is *proportional to the agent's own requirement* (more averse agents receive larger absolute reductions). This proportionality is the seed of the complementarity discussed in §5.5.

What it does not have.

- **No empirical calibration.** In both papers γ is a free "intensity" parameter, introduced with one sentence (2024: "for some innovations, social reinforcement may be more intense than for others ... so we introduce social reinforcement as a parameter") and *swept*: $\gamma \in [0, 1]$ continuously in the 2018 phase maps and the 2024 heat maps, with focal values $\{0, 0.5, 1\}$ in 2024's Fig. 5 ("$\gamma = 1$ is maximum social reinforcement"). There are no units, no measurement strategy, no experimental anchor — not even a link to Centola's dose–response evidence, which would have been the natural one.
- **No transportability across degree scales.** All Tur simulations live on $k = 4$ neighborhoods, so $a \in \{1, \ldots, 4\}$ in practice and the reinforcement spans at most a factor $4^\gamma \leq 4$. On NDFU's imputed networks (mean degree ≈ 29, hubs beyond 60), the same γ produces reductions up to 29^γ: at $\gamma = 1$ a saturated average neighborhood divides the requirement by ≈ 29, a force of a different order of magnitude. Even their $\gamma \leq 1$ bound is reasoned *for* $k = 4$. **Borrowing Tur's γ values would inherit the symbol, not the calibration.**

Verdict. Adopting the γ-form buys (i) exact formal nesting, (ii) the published axioms and the concavity argument for free, and (iii) a citation in the target journal. It does *not* buy empirical values for γ, nor freedom from arguing the coupling's magnitude — if the γ-form is used, γ must be calibrated for *our* degree distribution (e.g. with the same fit-to-current-model procedure used for λ). Epistemically, γ and λ are in the same place: free couplings — with the difference that λ has the stronger microeconomic backing (§5.6).

5.4 Exposure traditions: the power form is wedded to counts

The γ-form cannot be written with normalized exposure, and the reason is structural, not stylistic: the multiplier $\tilde{A}^\gamma$ amplifies only because $\tilde{A} \geq 1$. Substituting the fraction $\tilde{E} \in [0, 1]$ gives $\tilde{E}^\gamma \leq 1$ — social proof would *reduce* situational utility. There is no natural "fractional γ-form".

Consequences:

- Choosing the γ-form moves the model from the fractional-threshold tradition (Granovetter 1978; Valente 1996; Watts 2002) — the tradition NDFU and `netdiffuseR` live in — to the absolute-count tradition (Centola & Macy 2007). For a *Social Networks* audience this is not neutral.
- **Mitigation 1 (mechanism ≠ measurement):** even if the decision rule runs on counts, Valente-style exposure-at-adoption can still be computed and reported *ex post* as a descriptive statistic (measured thresholds). Nothing forbids reporting $\tilde{E}_i$ at the moment of adoption.
- **Mitigation 2 (tooling):** `netdiffuseR` supports non-normalized exposure (`exposure()` with `normalized = FALSE`), so the count threshold $\lceil (q_i/\Gamma)^{1/\gamma} \rceil$ runs on the same machinery. The loss is conceptual continuity, not implementation.

5.5 Are preferences modified? Composition, not units

Formally, the γ-form does *not* force the "preferences change" reading: $\Gamma \cdot \tilde{A}^\gamma \geq q_i$ is symmetric, and the factor $\tilde{A}^\gamma$ can be booked on either side —

$$\underbrace{U_i(t) = \Gamma \cdot \tilde{A}_i^\gamma \geq q_i}_{\text{situational utility multiplied; preference fixed}} \quad \Longleftrightarrow \quad \underbrace{\Gamma \geq q_i \cdot \tilde{A}_i^{-\gamma} = q_{i,t}}_{\text{requirement lowered (Tur's framing)}} \tag{18}$$

with identical behavior. In logs the first reading is a Cobb–Douglas-style utility in two "goods" (product quality, social proof): $\ln U_i = \ln \Gamma + \gamma \ln \tilde{A}_i$. Both sides live on the same $[0, 1]$ utility scale as q_i, so the real distinction is not one of *units* but of **composition**: additive vs. multiplicative.

And the composition carries substance. Compare the marginal value of social proof:

$$\text{linear:} \qquad U_i = \Gamma + \lambda \tilde{E}_i, \quad \frac{\partial U_i}{\partial \tilde{E}_i} = \lambda, \qquad \frac{\partial^2 U_i}{\partial \Gamma \, \partial \tilde{E}_i} = 0 \qquad (\textbf{substitutes}), \tag{19}$$

$$\gamma\text{-form:} \qquad U_i = \Gamma \cdot \tilde{A}_i^{\gamma}, \quad \frac{\partial U_i}{\partial \tilde{A}_i} = \gamma \, \Gamma \, \tilde{A}_i^{\gamma-1}, \quad \frac{\partial^2 U_i}{\partial \Gamma \, \partial \tilde{A}_i} = \gamma \tilde{A}_i^{\gamma-1} > 0 \quad (\textbf{complements}). \tag{20}$$

In the linear form, a given (similar) adopting neighborhood contributes the same $\lambda \tilde{E}$ regardless of how good the product is: quality and social proof are *substitutes* (additively separable), and the social contribution needed to convert agent i compensates the utility deficit one-for-one, $\lambda \tilde{E}_i \geq q_i - \Gamma$, independently of Γ's level. In the γ-form the cross-partial is strictly positive (complementarity in the Edgeworth–Pareto sense): *the same adopting neighborhood is worth more for better products*. The mirror statement on the requirement side is Tur 2018's own Eq. 2, where the drop is proportional to q_0^i. The required social proof falls steeply with quality,

$$\ln c_i^* \;=\; \frac{\ln q_i - \ln \Gamma}{\gamma}, \qquad \frac{d \ln c_i^*}{d \ln \Gamma} \;=\; -\frac{1}{\gamma}, \tag{21}$$

an elasticity of $-1/\gamma$ — e.g. at $\gamma = 0.5$, doubling the product's quality divides the required count of similar adopters by four.

So the honest summary of the user's point: it *can* be framed as preferences-fixed either way, but the multiplicative interaction is a substantive behavioral commitment (quality–social-proof complementarity), not just bookkeeping — and since Tur themselves present the form as requirement-lowering, adopting their exact form and citation invites the "preferences change" reading unless a paragraph defends the other bookkeeping.

5.6 The Brock–Durlauf microfoundation of the linear form

The linear variant is not ad hoc: it is the deterministic limit of the canonical discrete-choice-with-social-interactions model of Brock & Durlauf (2001, *Review of Economic Studies* 68(2):235–260). In their setup, agent i chooses a binary action and derives utility

$$V_i(\omega_i) \;=\; \underbrace{h_i(\omega_i)}_{\text{private utility}} \;+\; \underbrace{J \, \omega_i \, \bar{m}_i^e}_{\text{social utility (conformity)}} \;+\; \underbrace{\varepsilon_i(\omega_i)}_{\text{random shock}}, \tag{22}$$

where $\bar{m}_i^e$ is the expected mean choice of i's *reference group* and $J \geq 0$ is the social-interaction (conformity) coefficient. Three points of contact:

1. **Additive separability of private and social utility is the standard assumption** of that literature — exactly the structure of the linear rule. Mapping: the private net utility of adopting is $\Gamma - q_i$; the social term is $\lambda \tilde{E}_i$, with $\tilde{E}_i$ playing $\bar{m}_i^e$ (the adoption rate of the reference group) and λ playing J. Adoption when total utility is non-negative: $\Gamma - q_i + \lambda \tilde{E}_i \geq 0$ is (3) rearranged.

2. **NDFU's additions are well-defined deviations, not violations:** the reference group is not exogenous but *similarity-gated* (the $\sigma(h - d_{ij})$ filter selects who belongs to i's effective reference group — our contribution), and the shock is degenerate ($\varepsilon \to 0$), turning Brock–Durlauf's logit choice probabilities into a deterministic threshold rule. Reintroducing a small ε would give a smooth (logit) adoption rule — a natural robustness variant.

3. **J has an econometric life that γ lacks.** The Brock–Durlauf coefficient underpins the social-multiplier / social-interactions estimation literature (see Blume, Brock, Durlauf & Ioannides 2011, *Handbook of Social Economics*, for identification), so λ inherits a path toward empirical anchoring. As a bonus, Brock & Durlauf themselves develop the mean-field correspondence of (22) with the Ising model — which connects this microfoundation directly to NDFU's criticality analysis.

5.7 Verdict table and recommendation

	Linear-fractional — $\Gamma + \lambda\tilde{E}_i \geq q_i$	**γ-form (count)** — $\Gamma \cdot \tilde{A}_i^{\gamma} \geq q_i$
Answer to Bianchi	✓ equally bottom-up (threshold derived from measured q)	✓ equally bottom-up
Backing of the coupling	λ with Brock–Durlauf microfoundation (additive private + social utility; §5.6); its magnitude still needs arguing	γ with Tur precedent (axioms + concavity), but **no empirical calibration and no transportability** across degree scales (§5.3)
Nesting in Tur 2024	only qualitative	**exact, on any graph** (gate off; ξ=Γ, u_0=q, same γ)
Exposure tradition	✓ Valente/Granovetter fractional; `netdiffuseR` native	counts (Centola–Macy/Watts); Valente exposure only as ex-post reporting (§5.4)
Preferences-fixed framing	✓ clean ("situational utility rises"; Brock–Durlauf)	defensible, but against the grain of the very citation it borrows (§5.5)
Quality × social proof	substitutes (social contribution independent of Γ)	**complements** ($\partial^2 U/\partial\Gamma\,\partial\tilde{A} > 0$) — strong assumption, but interesting and testable
Immunity	$q_i > \Gamma + \lambda$, degree-independent (§5.1)	$q_i > \Gamma k_i^{\gamma}$, **degree-dependent** (averse + poorly connected) — arguably more realistic
Empirical status	✓ **already validated**: replicates the current model's heatmaps; $\lambda \approx 0.7$ identified unimodally	not yet run; γ to be calibrated with the same procedure
Hub behavior	bounded ($\tilde{E} \leq 1$)	at high γ, a saturated hub receives reinforcement $\sim k^{\gamma}$ — may over-fire on dense networks

Recommendation. Use the **linear-fractional variant as the primary operationalization** (validated, Valente-compatible, microeconomically clean, λ identified from the legacy data), and the **γ-form as the formal bridge to Tur and as a robustness check**: present the exact nesting (§4.6) in the text, and run a short γ sweep — calibrated with our own fit procedure, never by borrowing Tur's numbers — to show that the conclusions do not hinge on the additive-vs-multiplicative composition. This keeps the best of both: the tradition and the validation of the linear form, and the exact nesting as an argument, without inheriting γ's transportability trap.

6 Side-by-side with the current model and Tur

	NDFU current	Tur 2024	**Unified rule (both variants)**
Decision rule	two channels (OR)	single utility rule	**single utility rule**
Threshold source	exogenous $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$	emerges from reinforcement	**derived from measured** q_i: $(q_i - \Gamma)/\lambda$ or $(q_i/\Gamma)^{1/\gamma}$
Preferences modified during diffusion?	no	yes (u lowered)	**no** under the situational-utility framing (§4.2, §5.5)
Distance enters via	stochastic gate in $\tilde{E}$	— (absent)	**same stochastic gate**, in $\tilde{E}$ or in $\tilde{A}$
Heterogeneity source	q empirical + τ artificial	u artificial	q **empirical only**
Free params (beyond Γ, h)	τ_{mean}, τ_{sd}	γ	λ (linear) or γ (γ-form)
Nests Tur 2024?	no	—	**exactly via the γ-form** (§4.6); linear form only qualitatively
`netdiffuseR` as-is?	yes	no	**yes** (fractional or count `threshold.dist`)
Main risk	Bianchi critique	distance-blind	heatmaps shift with the τ–q coupling (checked: they don't, §7)

7 Calibration notes and empirical status

- Comparability anchor: the main-text configuration of the current engine is $\tau \sim \mathcal{N}(0.40, 0.12)$. Under the linear-fractional variant, $\tau_i^* = \max\{0, (q_i - \Gamma)/\lambda\}$, so the implied threshold distribution *depends on* Γ *and the empirical* q *distribution.* The sweep script reports, per (λ, Γ): mean and SD of τ^*, share of instigators ($\tau^* = 0$), and share of immunes ($\tau^* > 1$).
- **Empirical status (linear variant).** An initial minimal test (12 networks, coarse grid; script since removed) showed the qualitative phase structure survives. The extended sweep (`10_unified_model_sweep.R`: 96 GSS networks, the engine's full 41×13 grid, $\lambda \in \{0.3, 0.4, \ldots, 1.5\}$, random seeding, 665,184 simulations) found a clean **unimodal best fit to the current model at** $\lambda \approx 0.7$ ($r = 0.94$ total, 0.98 rational, 0.93 social against the $\tau \sim \mathcal{N}(0.40, 0.12)$ surface; $\lambda \in [0.6, 0.9]$ all $r > 0.91$). Known differences: tipping one MSP-grid step earlier, and the immunity ceiling at low λ (§5.1).
- On the λ range: $\{0.3, \ldots, 1.5\}$ was a pragmatic bracketing (the minimal test showed 0.5 and 1.0 straddling the current model and 2.0 clearly too strong); the sweep's unimodal peak with degradation at *both* flanks justifies the endpoints ex post. Theoretical landmark: $\lambda = 1$ is the **immunity boundary** — since $q \leq 1$, $\lambda \geq 1$ implies zero structurally-immune agents at every $\Gamma > 0$, while $\lambda < 1$ leaves an immune share $\Pr(q > \Gamma + \lambda)$. There is no theoretical bound at $\lambda = 1$ (λ is a utility contribution, like Brock–Durlauf's unbounded J); production runs use the calibrated band $[0.5, 0.9]$.
- Still pending: the mirror γ-form sweep (count threshold $\lceil (q_i/\Gamma)^{1/\gamma} \rceil$ on the same machinery, γ calibrated by the same fit procedure), the remaining seeding strategies, and the ATP/innovation system.

7.1 The structural premium survives the unified rule

Run by `scripts/05_unified_diffusion_sweep.R`: GSS vs. genuine degree-preserving twins (`igraph::rewire(keeping_degseq)`, built in memory with the recipe of `scripts/04_make_DP_networks.R`), $\lambda \in \{0.5, \ldots, 1.0\}$, 41 IUL levels × a 25-level MSP

grid (step 1/24, nesting the legacy 13-level grid), 96 networks, five seeding strategies (1,180,800 simulations each), with q_i = `mur_score` = 1 − propensity$_i$ (GSS mean 0.585). Analysis in `scripts/06_premium_sensitivity.R`.

Seed matching. The primary seed is drawn once per run, independently of λ. For *random*, *marginal* and *central* it is the *same node* in a GSS network and its DP twin; *closeness* and *eigen* use each topology's own most central node. The seed cluster is the primary plus its nearest nodes, 0.40× the network's maximum degree in total ($\approx$ 28 seeds), so strategies differ only in where they seed. The *random* strategy is the reference.

λ	0.5	0.6	0.7	0.8	0.9	1.0
β_{DP} (odds)	−0.482	−0.589	−0.753	−0.851	−0.922	−0.980
odds drop	−38%	−45%	−53%	−57%	−60%	−62%
$\bar{\Phi}_{\mathrm{GSS}} - \bar{\Phi}_{\mathrm{DP}}$ (pp)	4.0	4.5	5.5	6.0	6.3	6.6
% of plane with gap > 0.10	18.1	20.3	20.8	20.8	21.2	21.5
mean gap inside that band (pp)	16.5	17.6	21.3	23.4	24.6	25.8
MSP_c: GSS → DP	.085→.156	.088→.157	.082→.153	.079→.151	.078→.147	.077→.148

Reading. The premium survives the bottom-up engine across the whole λ band and for all five seeding strategies: rewiring the ties lowers the odds of adoption by 38–62% under random seeding (pooled $\beta_{\mathrm{DP}} = -0.684$, −50%). It grows smoothly and monotonically with λ (steps per decile −0.107, −0.164, −0.098, −0.071, −0.058): the stronger social proof is, the more the real fabric matters.

λ is a free coupling, not a calibrated constant, so the paper should report results as functions of λ. Two scales must be read together:

- on the **odds** scale the premium roughly doubles over the band (−0.48 → −0.98);
- on the **probability** scale it grows by +65% (4.0 → 6.6 pp).

Part of the difference is a link effect (cells with $\Phi > 0.95$ go from 40% to 61% of the plane for GSS, from 32% to 48% for DP, and the logit inflates a given probability gap there), but most of the growth is genuine structural separation.

The λ-robust statements (recommended for the paper):

1. **Where it lives.** The premium is concentrated in an L-shaped band along the tipping frontier: 18–22% of the (IUL, MSP) plane, where GSS gains 16–26 percentage points. Outside that band the two topologies agree (both dead, or both saturated). Structure decides where the system is undecided.
2. **Tipping shift.** DP needs 1.8–1.9 times the social openness to tip ($MSP_c \approx 0.077$–0.088 for GSS vs. 0.147–0.157 for DP). The absolute shift is the most stable way to state the premium: 0.069–0.071 across the band under random seeding (±2%), and 0.060–0.076 across all five strategies.
3. **Channel split.** The rational channel is flat in λ ($\beta \approx -0.146$) while the social channel carries the growth (−0.21 → −0.37). The rational channel is *not* topology-blind: its adopters still need one gate-passing contact, and DP ties are socially more distant (mean Gower 0.229 → 0.325), so the gate rejects more often.

Seeding strategies. Pooled and endpoint β_{DP}, the probability-scale gap and the tipping shift, by strategy (mean seed-cluster size in parentheses):

strategy	pooled β_{DP}	β_{DP}, $\lambda = 0.5$	β_{DP}, $\lambda = 1.0$	gap (pp)	MSP_c shift
central (28.7)	−0.758	−0.517	−1.179	4.0–6.9	.060–.068
closeness (28.0)	−0.757	−0.520	−1.175	4.0–6.8	.062–.076
eigen (28.0)	−0.747	−0.509	−1.169	4.0–6.8	.066–.073
random (28.4)	−0.684	−0.482	−0.980	4.0–6.6	.069–.071
marginal (28.4)	−0.685	−0.490	−0.955	4.0–6.4	.068–.074

- The three **hub strategies are interchangeable** (per-λ differences ≤ 0.02), even though closeness and eigen are not node-matched between GSS and DP: the hub premium is not an artefact of unequal seed budgets.
- **Random and marginal coincide** (pooled −0.684 vs. −0.685; run-level gap 5.46 vs. 5.44 pp): with an equal budget, a typical and a peripheral primary seed equally well.
- The **hubs add a small premium** (pooled −0.76 vs. −0.68) that grows with λ (at $\lambda = 1$: −1.18 vs. −0.97); run-level central − random = 0.15 pp, 95% CI $[0.02, 0.28]$.

A reverse-premium corner (conditional, not universal). In a small region — high openness ($h \gtrsim 0.6$, median 0.83) and moderate attractiveness ($\Gamma \approx 0.275$–0.45) — **DP beats GSS**. Counting cells where the gap is below −2 pp:

- the same in every strategy: at $\lambda = 0.5$–0.7 (2.6–3.4%, 2.4–2.6% and 0.8% of the plane, up to ≈7, 5.5 and 3 pp), gone at $\lambda \geq 0.8$ (≤ 0.3 pp).

The mechanism is Tur (2024)'s own: when the gate is wide open, homophily *isolates the resistant among the resistant*, whereas random mixing seats adopters next to reluctant agents and converts them. Once influence stops being similarity-gated, the sign of the structural effect can flip (the similarity-vs-diversity question). The corner appears only where $\Gamma + \lambda \lesssim 1$, i.e. where a structurally immune tail exists; this is a lead, not an established mechanism (it did not hold under the inverted q). The corner keeps the premium claim properly conditional, but it is a weak-coupling feature and should not be presented as present at every λ.

Immunity under the correct q. Agents with $q_i > \Gamma + \lambda$ can never adopt. At $\Gamma = 0$ they are 66, 47, 32, 16, 6 and 0% of agents for $\lambda = 0.5, \ldots, 1.0$; averaged over the IUL grid, 14.0, 8.1, 4.1, 1.7, 0.4 and 0%. Mean adoption over the plane is correspondingly lower (random seeding: $\bar{\Phi}_{\mathrm{GSS}} = 0.61$–$0.74$, $\bar{\Phi}_{\mathrm{DP}} = 0.57$–$0.68$).

What changed with respect to the inverted q. Until 2026-10-05 the engine read the propensity score itself as q_i (high = *more* willing), the opposite of its role as a minimum utility requirement (GSS mean 0.415 instead of 0.585); the same inversion affected the legacy engine. With the correct q: (i) the premium survives in every strategy and keeps its shape (L-shaped band, flat rational channel, stable tipping shift of ≈ 0.065–0.07); (ii) it is smaller on the odds scale (random pooled −0.750 → −0.684; hubs ≈ −0.93 → −0.75) but steeper in λ (random −0.61 → −0.96 became −0.48 → −0.98); (iii) the reverse corner moves to higher Γ (from 0.10–0.275, by about the 0.17 shift in mean q) and is no longer present at every λ; (iv) the immune share rises 2.7-fold or more (IUL-grid average 5.2 → 14.0% at $\lambda = 0.5$). The inverted-q outputs are archived in `output/**/_qinverted/` (closeness and eigen references: the pre-seed-fix runs in `_prefix/`, which match the seed-matched code for the deterministic strategies: central −0.926 vs. −0.928).

Still open. The BAM standard errors ($\sim 5 \times 10^{-4}$) treat every cell as independent and are far too optimistic: the unit of replication is the run (96 seed draws per topology), so the interval on β_{DP} must come from a run-level bootstrap (planned, with a joint BAM of the five strategies). Until then, differences between strategies of a few hundredths are not interpretable. The legacy reference ($\beta = -0.850$), the $\lambda \approx 0.7$ calibration (§7) and the immunity table in §5.1 were computed with the inverted q and are pending recomputation.

8 To verify before writing any of this into the paper

- ☒ Exact reinforcement functional form in Tur 2024: $u_t = u_0 \cdot (1/a_t)^\gamma$ (multiplicative, power-law). *Resolved against both PDFs (2026-08-31):* Tur 2018 (PRE), Eq. (1), uses the **same** form (there called MQR, $q_t^i = q_0^i(1/a_t^i)^\gamma$); $a_t^i \in \mathbb{N}$ is explicitly the number of **adopting neighbors** (a state count, not received recommendations), with the rule evaluated only once informed ($a \geq 1$). Both papers restrict $\gamma \in [0, 1]$ — justified in 2018 by non-increasing marginal social influence (concavity), with the reductio that large γ trivializes diffusion on their $k = 4$ lattices — and sweep γ as a free "intensity" parameter (2024, Fig. 5: $\gamma \in \{0, 0.5, 1\}$; heatmaps over $[0, 1]$). **No empirical calibration of γ exists in either paper**; see the transportability caveat in §5.3.
- ☐ Whether Tur 2024's homophily-rewiring ρ has a formal analogue we should cite when claiming nesting (their $\rho \approx$ our imputation; not identical constructs).
- ☐ The random-utility framing sentence for IUL (already drafted in `JOURNAL_FIT_ANALYSIS.md`, §4d) composes cleanly with the situational-utility framing: $U_i(t) = \Gamma + \lambda \tilde{E}_i$ is a private valuation plus a social term — this is Brock & Durlauf (2001)'s decomposition (§5.6), worth citing.